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Faculty of Systems Design Engineering

Canards Aerodynamic Design for an Active Roll-Control System



Waterloo Rocketry

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Dear PD evaluators,

This report, entitled Canards Aerodynamic Design for an Active Roll-Control System, is my preliminary draft for the final technical report of PD 11. I am currently completing my fourth co-op after successfully fulfilling my 2B academic term in Systems Design Engineering. I am working for the Waterloo Rocketry Design Team for which I am the flight dynamics lead. My supervisor is Tessa Pugh. The majority of work responsibilities on the team fall under the airframe and flight dynamics subsystems. Our design team bay is located in E5 1008 at the University of Waterloo. Waterloo Rocketry builds and designs sounding rockets in order to provide hands-on technical experience to undergraduate students. The flight dynamics subsystem is responsible for the aerodynamic design and simulation of the rocket to ensure its safety and targeted performance during launch.

This report describes the careful and methodical approach I have taken to the aerodynamic design of this year's active roll-control system. In the process, I am setting a precedent for future aerodynamic analysis on the team and providing necessary documentation for Launch Canada regarding the risks and mitigative steps taken to ensure the system's safety.

This report was written entirely by me and has not received any previous academic credit at this or any other institution. I have checked the report for spelling and grammar errors to the best of my ability. I would like to acknowledge the help of the Waterloo Rocketry flight dynamics and controls subsystems who helped develop and design this active controls system. I would like to specifically thank multiple team members whose work overlapped with mine in major ways. Richard Ding, Armaan Sengupta, Joe Dolina, Christine Li and Ben Pickens from the controls subsystem who helped define system requirements, identified risks and led the electrical, firmware and mechanical design of the canards. Finn Breuer for his contributions to the canards control algorithm. Naomi Weissberg for her work on coning analysis. Professor Jean-Pierre Hickey for his valuable insights and guidance in aerodynamics. And Joel Godard for his technical advice and help developing a rigorous approach to aerodynamic analysis. I am extremely grateful for the support provided by those mentioned above and many others I could not name.

Sincerely,

Luca Scavone
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Executive Summary

This report documents the design approach and aerodynamic analysis that led to the current iteration of Waterloo Rocketry's first active roll-control system. It focuses specifically on safety concerns and risks associated with their design and implementation on a liquid bi-propellant rocket targeting 50,000 ft in apogee and a maximum velocity of Mach 2.2.

This active control system will utilize two small delta-wing-shaped control surfaces referred to as canards. During flight, they will actuate in opposite directions, thus generating lift and inducing a rotational moment about the rocket's longitudinal axis. The canards have been sized such that they maintain control authority under 6Hz conditions for the first 30 seconds of flight.

Canard-associated safety concerns include positioning, coning, wake interference and flow separation. To mitigate these concerns, the canards are mechanically-linked and positioned well above the rocket's fins. Decreasing the risks associated with wake interference may come at a slight loss of stability, which will be compensated for with the implementation of a larger spin can. Canard actuation angle is also limited to 10 degrees, and incorporates a delta-wing shape design to reduce drag at supersonic speeds.

Concluding, a discussion on when the canard system should be activated is included, arguing that it is safer for the canards to be active upon launch, which contradicts a ruling from Launch Canada. The team will continue working on further analysis while waiting for a response from Launch Canada's judging panel.

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1 Introduction

Waterloo Rocketry is a student design team at the University of Waterloo that exists to provide students with opportunities to engage in hands-on learning through practical engineering work. Currently, around 100 students are involved in the team. Every year, the team designs, builds and launches at least one rocket and has competed in international and domestic rocketry competitions such as the Spaceport America Cup and the Launch Canada Challenge. Waterloo Rocketry prides itself on designing and manufacturing as many components in-house as possible. The team has many subsystems to accomplish this. Selected subsystems include propulsion, airframe, recovery, software, and the focus of this report: flight dynamics.

1.1 Team History

Waterloo Rocketry was founded in 2009 and has been actively participating in competitions since 2011. The team has historically focused on student-researched-and-designed (SRAD) hybrid rockets, but in 2024 made a significant transition to liquid bipropellant systems, achieving Canada's first liquid rocket launch with Borealis, achieving an apogee of 19,212 ft. This rocket was the team's most ambitious rocket in terms of technical scope, incorporating new complex propulsion, recovery and control systems. In particular, this rocket featured the team's first active-control system that utilized airbrakes to control the rocket's final apogee during flight. Unfortunately, the rocket was underfilled with propellant at launch and never reached an altitude high enough for the system to activate.

1.2 Launch Canada 2025

The team is competing in the 2025 Launch Canada Challenge with a high-altitude liquid bipropellant rocket named Aurora. The primary mission goal is a nominal ascent, followed by a successful recovery. The target apogee for this rocket is 50,000 ft, marking a significant

increase in targeted apogee over previous rockets that the team has flown. Building on Borealis' airbrakes system, this year's rocket also features roll control canards, marking a key advancement in the team's control capabilities. Given the flight conditions that Aurora is likely to experience, the aerodynamic design of the canards is of utmost importance to increase the safety of the system and will be the subject of the following report.

1.3 Flight Dynamics Subsystem

The flight dynamics subsystem was created in response to the increasing complexity of the team's projects, and the need for a dedicated branch to oversee simulations and aerodynamic design. Most notably, the subsystem introduced a cycle-based design system for LOTS (2023 Spaceport America Cup rocket) which led to a 23% increase in apogee by improving engine efficiency, system integration, dry mass, and much more. The team saw similar success in 2024 through the use of this system. The subsystem is currently led by Luca Scavone, the author of this paper.

The subsystem's recurring responsibilities include mass, length and CG tracking throughout the design process to inform general rocket architecture, fin design, and apogee prediction. As the team aims to go higher and faster, the subsystem's focus has expanded to include rocket optimization for high altitudes and aerodynamic support for current and future active control projects.

2 Selected Topics in Rocketry and Aerodynamics

2.1 What is a Liquid Bipropellant Sounding Rocket?

A liquid bipropellant sounding rocket, uses two separate liquid propellants—liquid ethanol as fuel and nitrous oxide as the oxidizer in the case of Waterloo Rocketry—that are mixed and burned in a combustion chamber to generate thrust. In contrast, solid rockets use a pre-mixed propellant that burns upon ignition, making them simpler to manufacture, whereas liquid rockets require complex plumbing, pumps, and precise control systems. Sounding rockets are designed to carry payloads on suborbital trajectories, reaching high altitudes for scientific research, but they are not intended to reach orbit.

2.2 Rocket Reference Frames

A rocket's motion is generally described using a three-axis reference frame centered around its center of gravity (CG), as shown in [Figure 1](#). The yaw axis runs horizontally through the rocket, perpendicular to its longitudinal axis, and controls left and right rotation. The pitch axis is also perpendicular to the longitudinal axis, governing the rocket's up and down tilting. The roll axis runs parallel through the rocket's body and determines the rotation around itself. These rotational motions—yaw, pitch, and roll—are fundamental in aerospace dynamics, as they define the rocket's orientation and stability throughout flight.

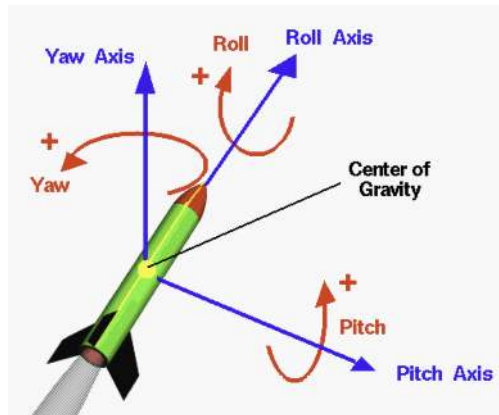


Figure 1: General rocket reference frame [\[1\]](#)

2.3 How Does a Rocket Stay Straight?

Figure 2 illustrates how a rocket maintains stability in flight by ensuring that its center of pressure (CP) is positioned behind its center of gravity (CG). The CP is the point where aerodynamic forces act, while the CG is the balance point of the rocket's mass. When the CP is behind the CG, any angular displacement from the flight direction generates a restoring aerodynamic force that pushes the rocket back to its intended path. This passive stability mechanism, aided by aerodynamic surfaces like fins, ensures that the rocket remains aligned with its flight direction rather than tumbling uncontrollably. The shape and size of the fins determines how far back the CP is positioned behind the CG.

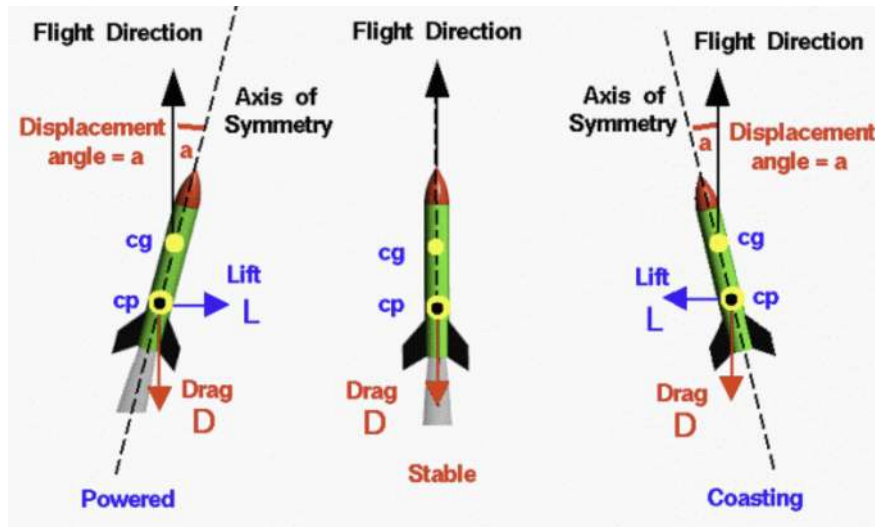


Figure 2: Rocket's remain stable by ensuring their center of pressure is positioned behind their center of gravity [1]

2.4 Introduction to Airflow

As a rocket is launched, it moves quickly through the atmosphere, efficiently pushing air out of the way due to an aerodynamic design. The shape of the rocket, typically a slender body with a pointed nose cone and stabilizing fins, minimizes drag and allows it to slice through the air with reduced resistance. The particles move smoothly out of the way in a flow described as laminar. As velocity increases, the airflow around the rocket becomes more complex, transitioning from subsonic to supersonic speeds. When the rocket travels at

speeds faster than the speed of sound, the particles cannot move out of the way fast enough and start to compress. This type of airflow is described as compressible. Furthermore, this compression of the air particles causes a shock wave to form, which resembles a cone of very thick air emanating from the tip of the rocket. The airflow may still remain laminar or may become turbulent, which generally increases drag and causes small rotational vortices in the airflow. The boundary layer, a thin region of airflow near the rocket's surface, also influences drag and heating.

2.5 How Does a Canard Work?

A canard quite simply describes a small wing positioned ahead of the rocket's center of gravity. A wing is a crucial aerodynamic control surface that uses its shape to create pressure differences between its top and bottom surface, thus creating lift. Generated lift is a function of air density ρ , velocity v , canard area A and lift coefficient C_L defined below in [Equation 1](#):

$$L = \frac{1}{2}C_L\rho v^2 A \quad (1)$$

Lift is defined as perpendicular to the neutral axis (relative wind vector), as shown in [Figure 3](#). Lift is also dependent on C_L which is a coefficient often defined in terms of the angle of attack α of the wing or canard. In the case of a canard, its symmetrical cross-section shape does not generate lift when $\alpha = 0$. Thus, for the canard to generate lift, it must be actuated to a non-zero angle of attack, which is precisely the role of the control system—it must adjust the canard's angle dynamically to achieve the desired aerodynamic effect.

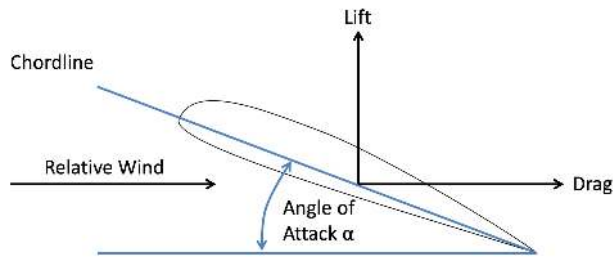


Figure 3: Lift is perpendicular to the neutral axis of the wing

3 Project Outline and Requirements

3.1 Aurora Design Goals

Building on the success of Borealis, the team is aiming to fly higher and faster this year, with a target apogee between 50,000 ft and 55,000 ft. The primary flight goal is to fly a nominal ascent and recovery with a significant amount of new innovative design elements. A secondary design goal is to develop systems and components that are designed for potential future flights to even higher altitudes.

3.2 Engine Thrust and Burn Time Targets

Initial engine performance targets focus largely on the off-rail velocity of the rocket. A general guideline for minimum off-rail velocity is 100 ft/s and a 20% safety margin gives 120 ft/s as the chosen target for off-rail velocity. Initial mass estimates for the vehicle were compiled based on previous mass estimates for Borealis and a breakdown of vehicle mass and length is included in [Figure 4](#).

Based on initial simulations, minimum thrust was set to 8 kN and a subsequent trade study between thrust and burn time was conducted to minimize propellant mass. Simulations were run both in OpenRocket to evaluate off-rail stability and velocity, and in RASAero for apogee prediction. Due to an unsuccessful static fire in November 2024, there are still several unknowns and uncertainties associated with the propulsion system that have prevented a specific thrust target from being set. Given this uncertainty, current engine thrust and burn time targets have been set to approximately 10 ± 1 kN with a 10-second burn time. Given current mass estimates, this corresponds to a thrust-to-weight ratio (TWR) equal to 12 and an off-rail velocity equal to approximately 150 ft/s.

2025 ROCKET LENGTHS			Mass Breakdown (lb)
Section	Sec. Length	Top plane distance	
Nosecone	40	0	10.44
Recovery	12	40	8.6
Avionics Bay	8	52	3.85
Canards	4	60	3
Feedsystem	72	64	50.64
Fuel tank	12	-	
Ox tank	58	-	
OTAS	4	-	
Engine Bay	16.5	135.50	19.18
CC	15.5	-	
DRY MASS		152.00	95.71
Fuel mass			15.87
Oxidizer mass			62.39
WET MASS			173.97
Length in ft		12.67	
Length L/D		19	

Figure 4: Aurora mass and length breakdown

3.3 Aurora Flight Conditions

Given the engine specification targets outlined in [subsection 3.2](#), Aurora’s maximum velocity is expected to reach Mach 2.2 and will remain supersonic for approximately 20 seconds. It will take Aurora 55 seconds to reach apogee and will experience a decrease in atmospheric pressure equal to 0.82 atmospheres. The rocket will also experience wind gusts and strong sustained winds at high altitudes caused by the jet stream.

3.4 Vehicle Breakdown

Aurora’s vehicle architecture borrows several features from previous rockets. As shown in [Figure 5](#), a conical nosecone (orange) will house a deployable payload to be ejected at apogee, while an upper body tube section (blue, green and purple) will house the recovery, avionics and canard systems respectively. Below the canards system, the fuel and oxidizer tanks make up the majority of the length of the rocket. The injector section (empty red section below the gray tank) houses the plumbing and electronics required to operate the propulsion system. The fin can (yellow) supports the rocket’s fins and houses the combustion chamber.



Figure 5: Aurora vehicle architecture

3.5 Canard Design Requirements and Considerations

The canard system being developed for Aurora is subject to several mechanical, aerodynamic, electrical and software design requirements established by both the team and the Launch Canada body. The primary requirements and considerations are listed along with their justification in [Table 1](#). Requirements are defined using the term “shall” while considerations and suggestions are defined using the term “should”.

Requirement	Justification
Canards shall be locked to explicitly roll control	The system will not control the yaw or pitch of the vehicle which presents significantly more technical challenges and safety risks
Canards should reduce the roll rate of the rocket to 0 deg/s	To avoid destructive coning behavior and produce better image quality
Canards shall not actuate (remain in neutral position) unless the rocket is at least 20,000 ft AGL, after boost phase or after flight has reached maxQ	As per Launch Canada's Design, Test & Evaluation Guide (DTEG) (Pending a successful test flight of this system in May, the team hopes to obtain permission from Launch Canada to activate the system from lift-off)
Each of the canard fins shall not operate independent of each other	No other moment shall be created other than roll moment on the flight vehicle
Canards shall be designed to take all the loading from directly impeding the airflow	The system must be strong enough to not break or fail in transonic and supersonic airflow; canards shall transmit all the moment and drag it induces to the rocket
Selected actuators shall be electrically driven	Generally simpler and more reliable
Canards shall not deploy when any of the rocket's yaw or pitch exceeds 30 degrees according to an IMU. It must abort and return to neutral state for the remainder of the flight	As per Launch Canada's DTEG
Canards should not induce roll reversal	Roll reversal can be destructive and could lead to the system losing control authority
Canards should be sized to mitigate roll in the region of flight most likely to experience roll	Due to the large altitude and velocity range during flight, the canards will not have control authority in some regions of flight
Canards should be shaped to minimize risks associated with parachute tangling upon drogue deployment at apogee	This could lead to a ballistic landing and total loss of the vehicle
Canards should be sized to minimize drag	This decreases expected loads on the system
Canards mounting mechanism should minimize protuberances on or near the control surfaces	This encourages non-turbulent airflow around the system

Table 1: Canard design requirements and considerations

4 Aerodynamic Approach and Methodology

While the goal of this report is to document the aerodynamic design of Aurora’s active roll-control system, a secondary goal is to develop a rigorous approach to aerodynamic studies for future use by Waterloo Rocketry. In 2024, the aerodynamic design of Borealis’ airbrakes system was disorganized and relied too heavily on computer-based analysis. Several key issues were identified after launch including:

- Communication breakdown between the controls and flight dynamics subsystems, resulting in disorganized timelines and a general lack of understanding of the various technical aspects by different project owners.
- Over-reliance on computational fluid dynamics (CFD) for aerodynamic analysis and aerodynamic properties that were crucial for the controller system to function properly. CFD is extremely time- and energy-intensive and requires reliable real-world data to verify and support its results. Notably, the system crashed several times during crucial simulations and compressible flow was omitted from the final simulation due to human error.
- Insufficient preliminary aerodynamic analysis proving the safety of the system due to decreased time commitments from the flight dynamics subsystem and no prior established methodologies within the team for conducting comprehensive aerodynamic studies.

To avoid a similar situation with Aurora’s roll-control system, key meetings in early September and October outlined subsystem roles and responsibilities, defined realistic scope and complexity, and established a rough outline of required aerodynamic analysis to demonstrate the system’s safety and to size the canards for the expected flight conditions. The following is a rough timeline of decisions and events throughout the development of Aurora’s roll-control system. The rest of the report is similarly structured according to the timeline.

1. **Initial design phase (September – early October 2024):** Initial design meetings defined technical scope for both the flight dynamics and controls subsystems; control authority was limited to the roll axis, mechanically-linked canards were suggested, and canards were tentatively positioned in the injector section.
2. **Early aerodynamic studies (October 2024):** Aerodynamic investigations began to assess expected flight conditions, focusing on key stability and control challenges:
 - Expected roll rate study to estimate the rocket’s maximum roll during flight.
 - Destructive coning study to understand pitch-roll coupling and identify contributing factors.
 - Roll reversal study to assess its causes, effects, and the potential need for a spin can as a mitigation strategy.
3. **Refinement of aerodynamic studies (end of October 2024):** The roll rate study was completed, and the spin can investigation evolved into a detailed wake interference analysis. This analysis focused on roll reversal issues caused by wing-tip vortices and shockwave interactions.
4. **Spin can requirement (end of November 2024):** The coning study was finalized; findings from the wake interference analysis confirmed the requirement of a spin can to mitigate roll reversal, regardless of canard placement on the rocket.
5. **Further aerodynamic analyses (December 2024):** Work began on calculating the lift generated by the canards and their induced angular acceleration.
6. **Repositioning of the roll-control system (January 2025):** In response to the wake interference findings and increasing integration challenges in the injector section, the roll-control system was relocated above the fuel tank to simplify assembly and reduce complexity.

7. **Finalizing canard design (mid-March 2025):** Lift and induced angular acceleration calculations were completed; canard shape was finalized. A preliminary design review was conducted to evaluate and refine the canard's size.
8. **Completion of canard sizing (mid-March 2025):** Following revisions from the design review, the canard sizing for both the Aurora rocket and its test flight vehicles was finalized.
9. **Future work (April-May 2025):** Minor aerodynamic studies including investigations on boundary layer formation, expected torque in the canard shaft, and pitch-induced stability risks in the advent of a mechanical failure.

5 Canard-Associated Safety Concerns

5.1 Positioning

The positioning of the canards on the rocket can greatly affect its stability. Two positions have been considered by the team: placing the canards above the fuel tank, or integrating them within the injector section below the oxidizer tank. The former offers a significantly easier design footprint to work with, but decreases the passive stability margin of the rocket by shifting the center of pressure (CP) forward and closer to the center of gravity (CG). This can be mitigated by increasing fin size to regain the loss in passive stability, but would lower their flutter velocity (this effect is likely negligible, but must still be considered). The latter option of placing the canards in the injector section is significantly more difficult from a system integration perspective, but does not affect the CP nearly as much since the canards are already located behind the CP. **Following further technical challenges, the team has decided to mount the canards above the fuel tank.**

5.2 Yaw and Pitch Control

To induce a rolling moment about the rocket's rotational axis, the canards located on opposite sides of the rocket must rotate in opposite directions. Doing so generates lift in opposite directions, which cancel, but produce a rotational moment without affecting the vehicle's yaw and pitch. However, in a scenario where the system were to malfunction or fail such that both canards actuate in the same direction, the lift forces generated would not cancel, which would cause a pitching moment about the vehicle's CG. This possible failure mode is of particular concern to the team, and ongoing research and aerodynamic analysis is being conducted to understand the full extent of its impact on the vehicle. This failure mode can be mitigated through a mechanically-linked canard design where the actuating shafts are connected such that rotating one canard causes an opposite rotation in the other. Furthermore, the axis of rotation of the canard can be placed above the expected CP of the canard, providing some

degree of passive stability to the canards and forcing them back to a more neutral position in the event where the actuating servo were to lose power. It is important to note that depending on the torque exerted by the servo, it is likely that the canards would not return to a fully neutral position.

5.3 Maximum Actuation Angle

Historically, the team has little experience manufacturing and designing lifting surfaces with airfoils. As such, the canards are expected to have an uncambered uniform cross-section with rounded leading and trailing edges. The rounded edges provide a smoother surface to deflect airflow while the uncambered cross-section ensures no lift is generated when the canards are in their neutral position (0-degree angle of attack). Furthermore, the canards can be modeled as flat plates that only produce lift when positioned at a given angle of attack (AoA) relative to the oncoming airflow. According to flat plate theory, wake separation occurs around an AoA of 12 degrees, resulting in a sharp drop-off in lift being generated. This is a significant concern from an aerodynamic standpoint that would also result in excess drag, turbulent airflow and potential loss of control for the controller. While this can be mitigated by implementing a maximum actuation angle, this limit comes at a loss of precision for the controller since the number of steps that the servo can make is decreased. For a flat plate in normal conditions, lift is proportional to AoA and canard surface area. Thus, for a canard limited to a 5-degree actuation angle, the canard itself must be quite large to produce the same lift as a canard limited to a 10-degree AoA with a smaller area. Furthermore, a 5 degree limit results in half as many servo steps, each step producing a larger difference in lift from the previous due to the canard's larger size. **The finalized design has been limited to a 10-degree actuation angle in either direction, with a mechanical stop at 12 degrees.**

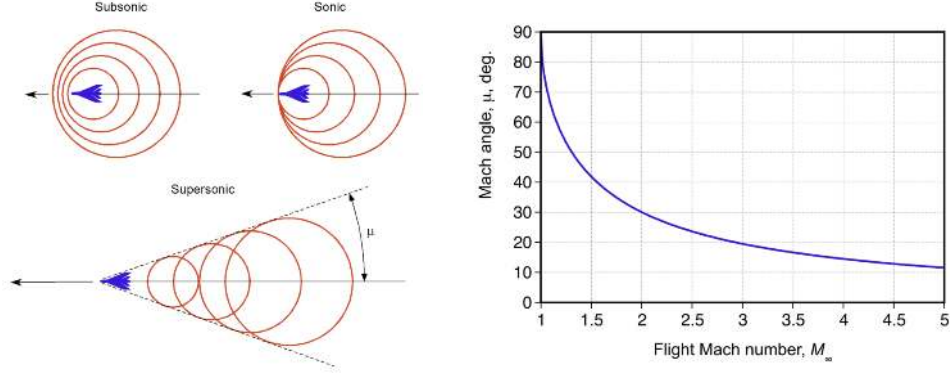


Figure 6: Evolution of the Mach cone half angle μ as a function of Mach number [2]

5.4 Canard Size and Shape

As briefly discussed in the previous subsection, the canards are likely to have a flat plate cross-section with rounded leading and trailing edges. With regard to the planform shape of the canards, the team is exploring swept wing designs that minimize drag at supersonic speeds. As the rocket reaches supersonic speeds, shockwave cones are likely to form around the canards. This cone extends outwards with a half angle μ as a function of mach number M according to Equation 2. Figure 6 illustrates the behavior of the Mach cone as a function of Mach number.

$$\mu = \arcsin \frac{1}{M} \quad (2)$$

For a swept canard with sweep angle Λ as shown in Figure 7, significantly more drag is created if the leading edge intersects the Mach cone when the vehicle is supersonic (i.e. when $90 - \Lambda > \mu$). As the Mach number increases, μ gradually decreases causing the cone to become sharper. Thus, choosing a design that ensures the canards are permanently swept behind the shockwave (referred as a subsonic leading edge) is important to decrease the expected loads on the canards caused by wave drag in the transonic region. A subsonic leading edge also results in smoother airflow over the canards which helps create more lift. These design features help address concerns about failure modes where one or both canards

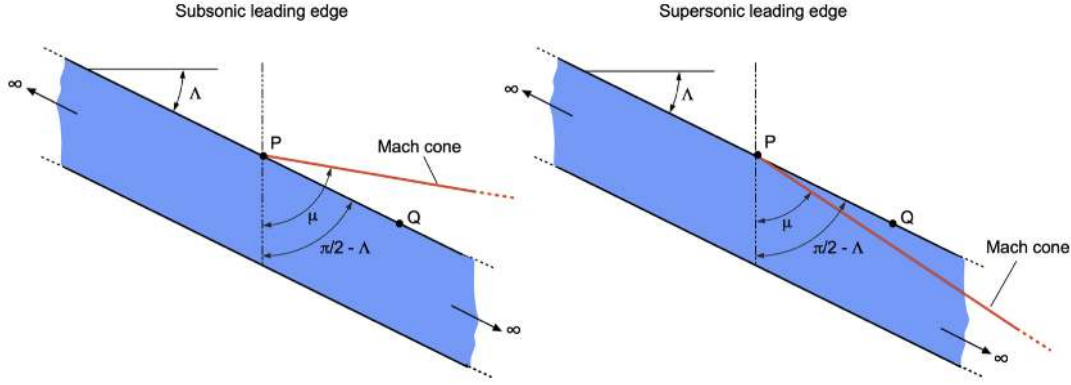


Figure 7: A swept wing with subsonic and supersonic leading edges. Limiting the buildup of wave drag on the wing requires that the leading edge be swept behind the Mach cone [2]

could be ripped off during flight due to unexpectedly high loads caused by wave drag. Flutter is also an area of concern given that the canard's root chord is not attached to the rocket's body tube. Further flutter analysis is being conducted on both the canard itself and the axis of rotation to determine mechanical manufacturing constraints in terms of minimum axle and flutter thickness.

5.5 Canard Wake Interference and Roll Reversal

Lifting surfaces generate lift by creating pressure differences between the top and bottom surfaces, disturbing the airflow in the process. The disturbances extend behind the lifting surface's trailing edge in the form of wakes which can interfere with other surfaces. Canard wake interference is likely caused by wingtip vortices, shown in Figure 8. While these vortices are relatively small just behind the canards, they can quickly grow in size and affect the stability of the rocket if they intersect any other control surfaces. Wake interference can also be caused by shock waves that asymmetrically intersect the rocket. This occurs when the canards are positioned at an angle of attack relative to the airflow, resulting in shockwave-expansion-fan pairs at the leading and trailing edges, illustrated in Figure 9. If the canards are positioned too closely to other control surfaces, the asymmetrical shockwaves cause uneven pressure distributions on said surfaces. Both shock waves and wing tip vortices may result

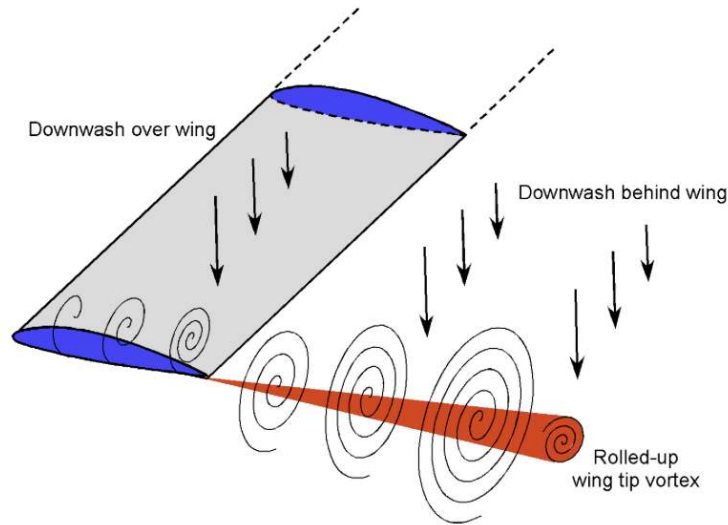


Figure 8: Wing tip vortices extend and grow in size behind the wing [2]

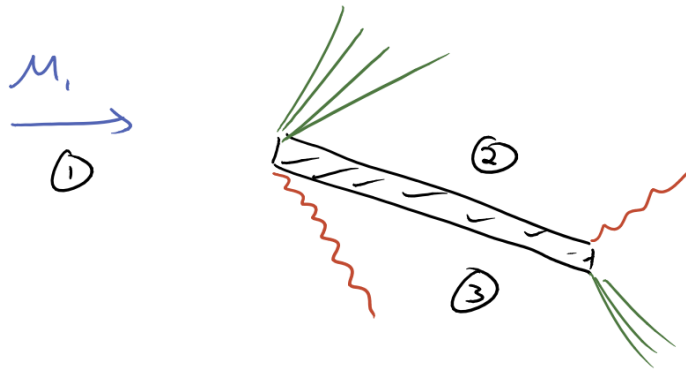


Figure 9: A flat plate at an angle of attack creates shockwave-expansion-fan pairs at the wing's leading and trailing edges

in roll reversal, an aerodynamic phenomenon that causes the rocket to rotate in the opposite direction than what is commanded. An extensive literature review conducted by the team revealed that no amateur rocket has successfully flown a canard roll-control system without additional mitigative measures such as a spin can (fin can placed on ball bearings allowing the fins to rotate freely without causing the rest of the rocket to roll).

6 Estimated Roll Rate and Coning

Aurora’s increased speed and targeted apogee could also increase the amount of roll that the rocket is expected to experience. Rockets experience roll due to small manufacturing defects in a rocket’s fins and/or external factors such as wind and angle of attack.

6.1 Retrospective Roll Rate Analysis

The size of the canards will be affected by the maximum expected roll rate the rocket could experience. A survey of other high-altitude rocket launches has shown that most nominal launches experience roll rates between 0.5 and 2 Hz. Based on an analysis of on-board video from past Waterloo Rocketry projects, LOTS experienced a maximum roll rate around 1 Hz during boost which decreased to approximately 0.67 Hz by burnout, while Borealis experienced a steady 0.2 Hz roll rate over the first 10 seconds of flight which was exceptionally low due to its high passive stability margin.

For the purposes of sizing canards, a factor of safety equal to 3 is applied to the 2 Hz upper bound determined previously. This is due to uncertainty associated with launching a rocket into the jet stream, potential fin misalignment during manufacturing, and the risk that the spin can may not function as intended. **Thus, when the canards are set to their maximum actuation angle, they must be able to spin the rocket to 6 Hz in approximately 0.5 seconds under MaxQ conditions** (an arbitrarily set time interval that ensures that the canards have sufficient control authority through the first 30 seconds of flight). This high factor of safety ensures that the canards maintain control authority under all flight conditions, including any unexpected aerodynamic behaviors the rocket may experience.

6.2 Coning Analysis and Mitigation

Coning is a destructive aerodynamic phenomenon caused by the coupling of roll, pitch, and yaw oscillations, which can occur when the rocket’s roll rate aligns with its natural pitch frequency. As Aurora is expected to exceed Mach 2.0, this risk has become a concern. To mitigate the potential for coning, an analysis of roll rate and pitch rate was conducted. An extensive literature review, including NASA reports, articles by rocketry companies and outside university rocketry projects, also revealed the following factors as potential contributors to coning occurrences:

- Fin misalignment
- Low number of fins
- Aerodynamic imbalances (from non-uniform fins or external wind)
- Flexible rocket sections and joints
- Uneven propellant burning
- Large LD ratios
- Inertial spin axis being displaced from the projectile center-line
- Center-of-gravity offsets

The team used mass estimates from Cycle 2, OpenRocket simulations and formulas [4] [3] to estimate the natural pitch frequency and roll rate. The pitch frequency was calculated according to Equation 3 below:

$$w_n = \sqrt{\frac{0.5pV^2 A_r C_{Na} [C_p - C_g]}{I_L}} \quad (3)$$

Where: w_n = natural frequency

p = density of air

V = velocity of rocket

A_r = reference area

C_{Na} = normal force coefficient

C_p = center of pressure location

C_g = center of gravity location

I_L = longitudinal moment of inertia

As per [subsection 6.1](#), during the boost phase, roll rate could be between 0.5 and 2 Hz, with a pitch frequency ranging from 0.0058 Hz at takeoff to 0.055 Hz at burnout. With the potential implementation of a spin can, the roll rate is expected to decrease further, significantly reducing the likelihood of coning. These results suggest coning is unlikely to occur during flight. Nonetheless, mitigative steps have been taken through the implementation of a spin can, previously mentioned in [subsection 5.5](#). A spin can places the fin can on ball bearings that allow it to rotate freely about the axis of the rocket to prevent the rest of the rocket from rotating. While the analysis above in [subsection 6.2](#) indicates the rocket is unlikely to experience coning, the addition of roll-control canards forward of the fins will result in wake interference known as roll reversal, an issue described in [subsection 5.5](#). Thus, the team has chosen to pursue a spin can design that will mitigate unwanted roll behavior from both the canards and any external factors such as sustained winds or wind gusts.

7 Calculating Lift

When the canards are actuated to some angle of attack, they produce lift perpendicular to the longitudinal axis of the rocket. This moment induces an angular acceleration to counter any roll the rocket is experiencing. Recall that generated lift is a function of air density ρ , velocity v , canard area A and lift coefficient C_L defined in [Equation 1](#).

7.1 Subsonic Flow

Accurately calculating the lift produced by the canards is important for canard sizing. Specifically, the team conducted extensive research to determine a decently accurate C_L value for the canards such that this sizing can occur. It should be stated that the controller itself does not specifically reference this lift coefficient during flight and will determine its own approximations. As such, the focus is not on determining the most accurate C_L value, but rather on a sufficiently accurate value that captures the aerodynamic behavior of the canard through subsonic, transonic and supersonic regions of flight as a function of angle of attack α . As the canards are uncambered, early research pointed to C_L models for flat plates derived from flat plate airfoil theory and linearized subsonic flow to account for the compressibility effects of the atmosphere as velocity approaches Mach 1.0. [Equation 4](#) describes C_L for subsonic compressible flow [\[2\]](#):

$$C_L = \frac{2\pi}{\sqrt{1 - M^2}}\alpha \tag{4}$$

Given the nature of the Prandtl-Glauert factor $\sqrt{1 - M^2}$ in [Equation 4](#), this results in an asymptote at Mach 1. However, given the swept nature of the canards, flat plate theory is not necessarily accurate. Stengel’s textbook on flight dynamics provides alternatives in [Equation 5](#), [Equation 6](#) and [Equation 7](#) that account for both aspect ratio AR and compressibility effects [\[6\]](#).

$$C_L = \frac{\pi AR}{2} \alpha \quad (5)$$

for $AR < 1$ according to slender-wing theory for any Mach number

$$C_L = \frac{\pi AR}{1 + \sqrt{1 + (\frac{AR}{2} \cos \Lambda_{1/4})^2 (1 - M^2 \cos \Lambda_{1/4})^2}} \alpha \quad (6)$$

for $M < 1$ according to Hembold's equation for subsonic flow

where $\Lambda_{1/4}$ is the sweep angle at the canard's quarter-chord

$$C_L = \frac{1.8 + AR}{1.8 + \sqrt{1 - M^2} AR} \cdot \frac{5.65}{1 + \frac{1.8}{AR}} \cdot \alpha \quad (7)$$

for $M < 1$ according to Prandtl-Glauert-Gothert-Ackeret similarity rules [5]

7.2 Supersonic Flow

Regarding supersonic flight, thin airfoil theory predicts lower C_L values that decrease as a function of the supersonic Prandtl-Glauert factor according to Equation 8 below. This equation however assumes a supersonic leading edge, which the team has decided against in the interest of reducing wave drag on the canards [6].

$$C_L = \frac{4}{\sqrt{M^2 - 1}} \alpha \quad (8)$$

For a subsonic leading edge, the supersonic flow over the canard often behaves more similarly to subsonic flow due to the independence principle that decomposes the velocity

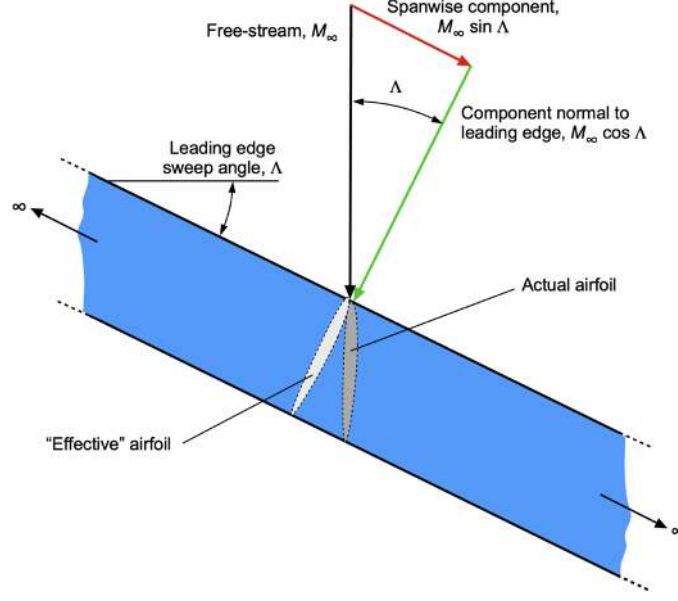


Figure 10: The independence principle states that only the Mach number acting normal to the leading edge affects the aerodynamics [2]

vector of the airflow into span wise and chord wise components, as shown in Figure 10. If the chord wise component of Mach number is smaller than 1, the canard is described as having a subsonic leading edge. Again, Stengel provides Equation 9 derived empirically for C_L as a function of angle of attack, sweep angle and Mach cone half-angle [6]:

$$C_L = \frac{2\pi^2 \cot \Lambda}{\pi + a} \alpha \quad (9)$$

$$\text{where } a = m(0.38 + 2.26m - 0.86m^2)$$

$$m = \frac{\cot \Lambda}{\cot \sigma}$$

$$\sigma = \arccos \frac{1}{M}$$

It should be noted that Equation 9 does not explicitly account for aspect ratio or compressibility effects. However, the equation implicitly accounts for some compressibility effects through the half-angle of the Mach cone σ and the sweep angle Λ , which influence the aerodynamic load. Since delta wings have a high sweep angle and low aspect ratio, the flow

generally remains more attached, and shock-induced separation is less significant, reducing the overall impact of compressibility effects. The team has chosen to size the canards according to this subsonic leading edge equation because it is continuous along the Mach number range Aurora is expected to fly in. Specifically, it does not produce the asymptote at Mach 1 which is the case for several of the equations presented earlier. [Figure 11](#) plots [Equation 9](#) for delta wings with varying sweep angles across multiple Mach numbers. As can be seen, this equation is also valid for subsonic flow where $\sigma = 0$. It is also continuous with [Equation 8](#) that predicts C_L for a supersonic leading edge. No other combination of equations that the team has investigated has provided a smooth transition across the subsonic, subsonic leading edge, and supersonic leading edge regimes. Other equations the team investigated, plotted in [Figure 12](#), illustrate the discontinuities or abrupt changes in predicted C_L when transitioning between these flow regimes, making them less reliable.

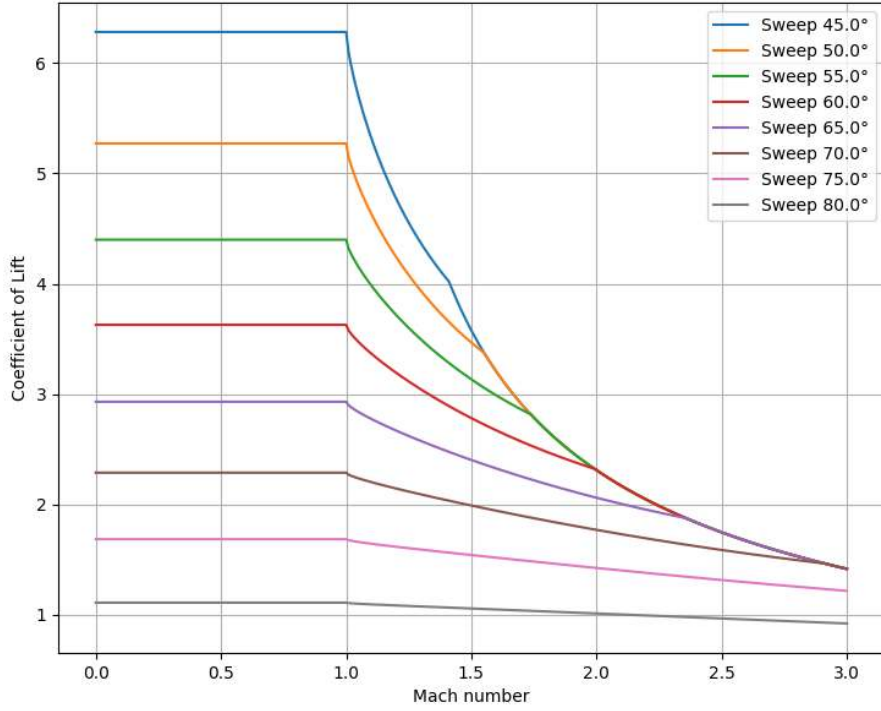


Figure 11: C_L as a function of Mach number and sweep angle according to [Equation 9](#)

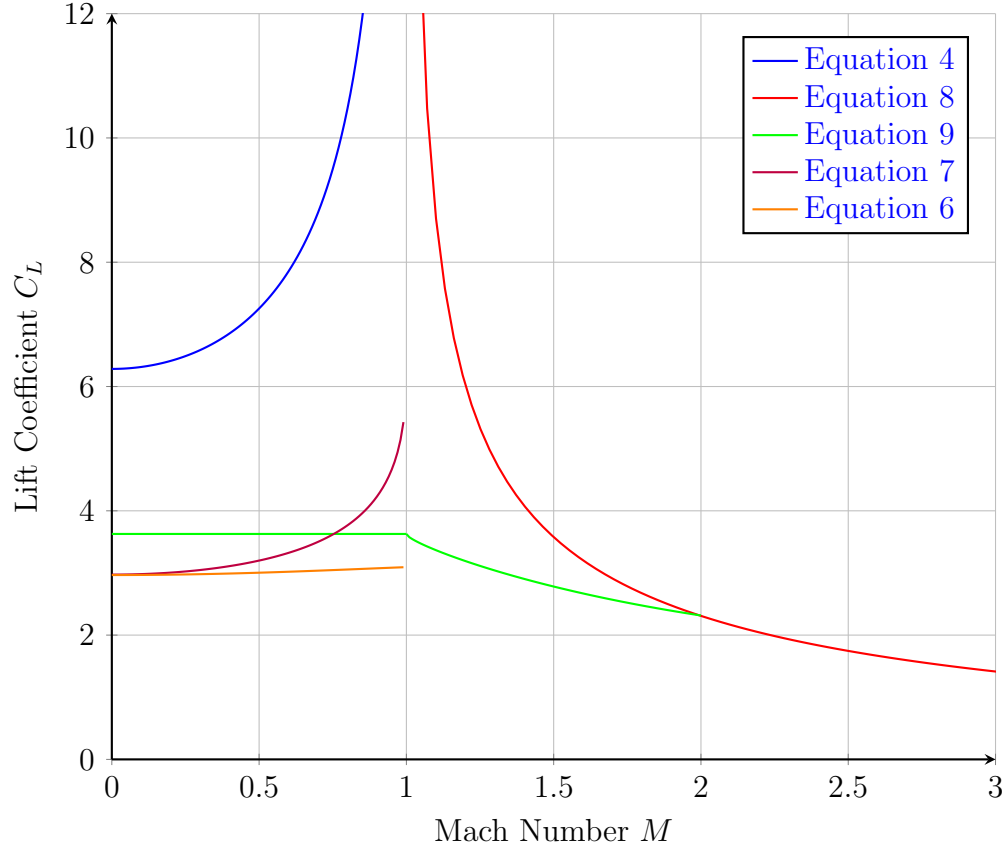


Figure 12: Comparison of lift coefficient models as a function of Mach number

The limit to Equation 9 is its behavior at subsonic speeds. It is widely accepted that the lift coefficient is at least partially influenced by compressible flow as the flow approaches Mach 1. Since Equation 9 is constant for any velocity below Mach 1, this constitutes a small inaccuracy. Thus, to model the lift created by the canards, the team chose a hybrid approach, utilizing Equation 6 for subsonic flow, Equation 9 for supersonic flow with a subsonic leading edge, and Equation 8 for supersonic flow with a supersonic leading edge. Lastly, while a discontinuity is still present at Mach 1, as shown in Figure 12, the difference was considered acceptable given the relatively short time frame the rocket spends traveling at speeds around Mach 1.

8 Sizing Canards

8.1 Current Design Configuration

The team’s initial configuration placed mechanically-linked canards below the CP in the injector section. While this design created significant integration challenges, it mitigated negative impacts from the canards on passive stability. It also reduced risks associated with both canards pitching in the same direction. Furthermore, the team was still analyzing wake interference and believed that placing the canards closer to the fins would not require a spin can.

Following further analysis of the behavior of the canards at supersonic speeds, it was determined that a spin can would be required no matter where the canards were to be positioned. While repositioning the canards above the fuel tank would require further aerodynamic analysis to prove their safety (given the negative stability margin associated with this location), the integration challenges associated with mechanical linkages in the injector section ultimately proved to be the deciding factor. The team has since opted for a mechanically-linked canard design positioned above the fuel tank. The fins will have rounded leading and trailing edges with a swept delta wing design that ensures a subsonic leading edge to reduce wave drag effects. The maximum actuation angle has been set to 10 degrees, thus providing more precision to the controller.

8.2 Sizing and Control Authority

Using trajectory data from an OpenRocket simulation of Aurora (time step, altitude, velocity, Mach number and atmospheric density), the estimated lift produced by the canards can be plotted as a function of time. From Cycle 2 Aurora mass and CG estimates, an approximate value for the rotational moment of inertia of the rocket, I , was determined allowing the resulting angular acceleration of the vehicle, $\dot{\omega}$, to also be calculated according to [Equation 10](#) below. The parameters in [Figure 13](#) were iteratively modified until [Figure 14](#) was obtained,

which illustrates the expected duration of flight where the canards are expected to have meaningful control authority.

$$M_x = L \cdot d = I \cdot \dot{\omega}$$

$$\Rightarrow \dot{\omega} = \frac{L \cdot d}{I} \quad (10)$$

Parameter	Value	Units
Angle of attack	10	deg
Angle of attack (rad)	0.1745	rad
Chord (in)	3	in
Span (in)	3	in
Surface area	4.5	in ²
Aspect ratio	2	
Lambda (rad)	1.107	rad
Lambda (deg)	63.426	deg
Surface area (m ²)	0.002903	m ²
Number of canards	2	
Rotational moment of inertia (dry mass)	1078.236	lb-in ²
Canard distance from center	4.5	in
Canard distance from center (m)	0.1143	m

Figure 13: Modifiable sizing parameters

Using basic kinematics, the time it takes to accelerate a rocket to a given roll rate can be calculated. To minimize safety risks but ensure sufficient control authority, the canards were sized by calculating this time-to-roll-rate value for the rocket at MaxQ (highest value in [Figure 14](#)) assuming full canard actuation to 10 degrees. Given how quickly the rocket accelerates and decelerates, this is by far the worst-case scenario in which the canards must maintain control of the vehicle. The rocket also reaches MaxQ at motor burnout, corresponding to the region of flight where the rocket is most likely to experience roll. As shown

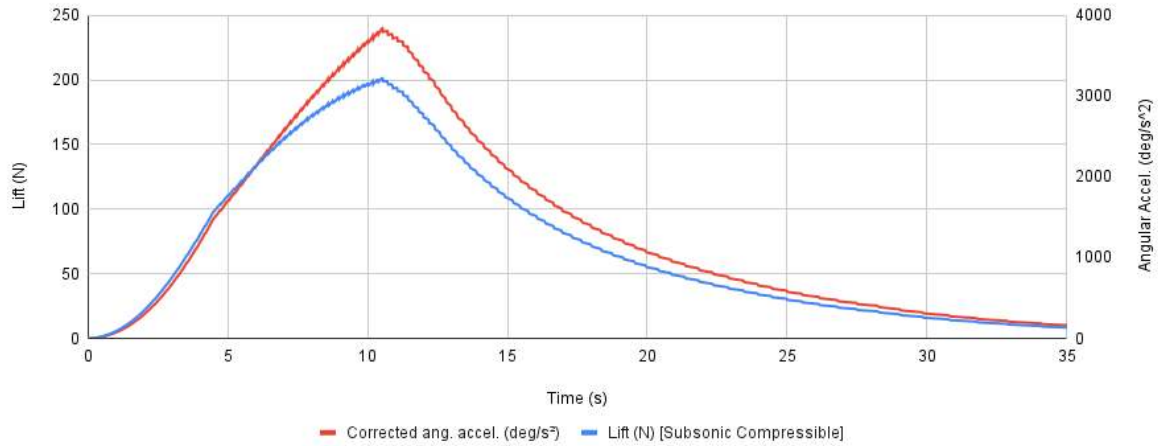


Figure 14: Predicted lift produced by the canards and resulting angular acceleration of the rocket as a function of time

in Figure 15, an average time interval was also computed to provide further context into average expected control authority over the first 30 seconds of flight (the atmosphere is too thin and the rocket is not traveling fast enough to maintain control authority after the first 30 seconds of flight). The parameters from Figure 13 result in a minimum time-to-roll-rate value of 0.562s and an average value of 1s. At first glance, this amount of control authority may seem excessive, but considering the nominal case where the canards are actuating by just a few degrees at a time, this time-to-roll-rate value quickly increases to over 5 seconds on average.

Parameter	Value	Units
Given roll rate	6	Hz
Angle of attack	10	deg
Choose angular acceleration (AVG)	2073.4005	deg/s ²
Time to dampen roll rate	1.042	s
Choose angular acceleration (MAX)	3840.13	deg/s ²
Time to dampen roll rate	0.562	s

Figure 15: Time-to-roll-rate minimum and average values

Following a design review where the aforementioned control authority was approved in-

ternally by the team, the following four designs were proposed in Figure 16. The first design furthest on the left is considered the default delta-wing case while the other three are variations that modify the tip chord sweep to satisfy the requirement outlined in subsection 3.5 that the canards should be shaped to minimize the risks associated with parachute tangling upon drogue deployment at apogee. By gently slanting the trailing edge, the parachute rigging is less likely to get caught on the canards in the event that the rocket tips over before the recovery system is deployed. The centroid and 60% root-chord-line are also indicated for each design. Due to flutter risks associated with longer root-chord-to-shaft ratios, the team selected the second design moving forward. However, this design slightly increases the expected drag and torque on the actuating shaft due to the smaller sweep angle. This was deemed an acceptable compromise, as the canard would experience a supersonic leading edge only for a couple of seconds at most.

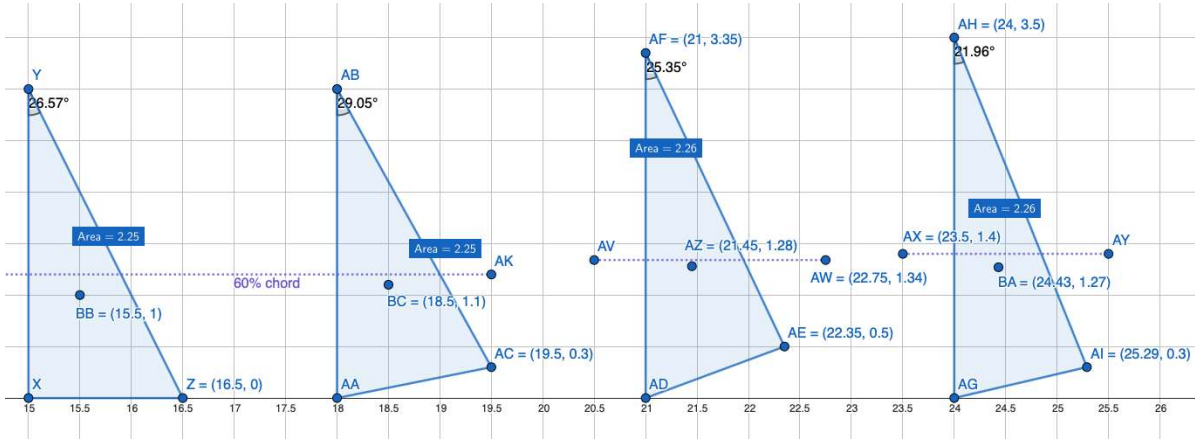


Figure 16: Proposed canard designs

8.3 Canard Activation

Canard sizing is heavily dependent on the region of flight where the rocket is most likely to encounter high roll rates. Thus, the canards must be capable of generating sufficient lift to mitigate the maximum roll rate the rocket is expected to experience during the region of flight where this occurs. Given the significant variation in rocket velocity and atmospheric

density throughout flight, the canards are not expected to have control authority over the entire flight.

Following an analysis of maximum roll rates experienced by other high-altitude hybrid and solid rockets, high roll rates most commonly occur during the boost phase coinciding with regions of maximum dynamic pressure and maximum velocity. It follows that sizing canards for this flight regime requires the smallest planform area, as dynamic pressure is at its peak. The team believes that if the canards are designed to provide sufficient control authority during the most intense phases of flight, their overall aerodynamic impact throughout the entire trajectory is minimized. Additionally, in the event of a failure mode where the canards negatively affect stability or aerodynamic performance, their reduced size helps mitigate adverse effects by minimizing their influence on the overall vehicle dynamics, reducing the potential for destabilizing moments or excessive drag.

The team's intent is therefore to have the canard control system active upon launch. This goes against Launch Canada's DTEG R7.4.1 that "CAS or ACS shall mechanically lock in a neutral state until the mission's boost phase has ended (i.e., all propulsive stages have ceased producing thrust)". Pending a successful test flight of the system, the team hopes to appeal the requirement by demonstrating that early activation does not compromise safety.

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